

WELL COMPLETION REPORT (SYNTHETIC)

OIL-RDS-004 - Rudrasagar-004

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000042
Field / District	Rudrasagar / Sivasagar
Block	PEL-ASSAM-06
Basin	Assam-Arakan
Formation	Kopili (KOP)
Spud / Completion	2011-06-28 / 2011-12-20
Measured / TVD	3418 m / 3366 m
Trajectory	Vertical; inc 1.1 deg; az 315.6 deg
Formation Interval	1830-3071 m
Lithology	Calcareous shale and siltstone
Pore Pressure Reference	4212 psi
Fracture Gradient	15.24 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	High (57.9/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	854	Surface
9 5/8 in casing	1982	Intermediate
7 in liner/casing	3007	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2011-07-27 724 m	Pressure Spike	High	Pressure trend exceeded recent moving baseline. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2011-07-29 771 m	Pressure Spike	High	Circulating pressure climbed without equivalent torque change. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Correlate pressure anomalies with flow before escalating risk.
2011-08-20 1186 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Track flow balance and pit volume trend before losses become severe.
2011-08-25 1299 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2011-09-11 1609 m	Torque Spike	High	Rotary speed fluctuated during constant pump rate. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2011-09-11 1620 m	Mud Loss	High	Active pit volume declined during stable pump rate. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2011-10-23 2386 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.

Completion Summary

The synthetic well reached 3418 m MD in the Kopili interval. The NWIS knowledge record links 16 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.